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JAMES WEBB TELESCOPE

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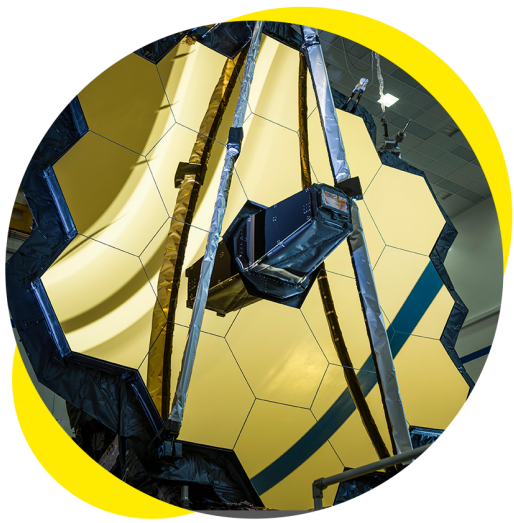
The James Webb Space Telescope is an orbiting infrared observatory. It is the world's most powerful' telescope' and on January 25 ,it reached the point called second Lagrange Point (L2) between Earth and Sun, which is 15,00,000 km away from Earth –more than four times as distant as the moon.

It has entered orbit here and is expected to work for a decade or two, watching the birth of our universe.The James Webb Telescope was launched from Europe's Spaceport in French Guiana on board the powerful Ariane - 5 rocket on Christmas day last year. JWST traveled at 843.12 kilometers per hour over a month to arrive at its destination.

Now that it has reached orbit, the James Webb Telescope will enter a controlled cooldown period which would allow it to better sense infrared information.It will also go through another five months of commissioning before it begins observing the universe in infrared. So, the first images from Webb can be expected to start coming in around five months.

The JWST will face the darkside of Earth, away from Sun,and observe the early universe going As far back as the big bang itself. It is set to solve mysteries about the solar system,look at distant stars and worlds and probe the mysterious structures and origins of our universe.

The mission goals of the James Webb Space Telescope are:



1.Search for the first galaxies or luminous objects formed after the Big Bang.
2.Determine how galaxies evolved from their formation until now
3.Observe the formation of stars from the first stages to the formation of planetary systems
4.Measure the physical and chemical properties of planetary systems, including our own Solar System, and investigate the potential for life in those systems

Facts about the James Webb Space Telescope

1.The JWST was launched on December 25, 2021 through the launch vehicle Ariane 5ECA. The total payload was approximately 6200 kg, including observatory, on-orbit consumables and launch vehicle adapter.
2.The golden mirror is the most incredible facet of the world's most powerful telescope. The diameter of the primary mirror is approximately 6.5m (21.3ft). The material of the primary mirror is beryllium coated with gold and it weighs 705 kg.
3.The focal length of the mirror is 131.4 meters. The primary mirror has 18 segments.